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Supply Chain Optimization using IoT and Machine Learning for Inventory Classification.

YASMEEN SHAIKH, ALIYA KILLEDAR, MISBAH DESAI AND SHREYA HOMBALKAR

Department of Computer Science and Engineering, S G Balekundri Institute of Technology, Belagavi 590010, Karnataka, India

Corresponding author: Prof. Yasmeen Shaikh (y.s.shaikh1210@gmail.com)

Abstract:*The paper focuses on supply chain optimization through the integration of Internet of Things and Machine Learning for inventory management, demand forecasting, and automated report* ***generation. It also BM quantum computers leverage furniture a comprehensive database of medicine*** *details, incorporating uses, side effects, dosage instructions, and relevant precautions. The inventory management system comprises of RFID and NODEMCU for efficient tracking, and environmental temperature monitoring is achieved through temperature sensors. The entire system is built using Django, Python, HTML, CSS, Vanilla JS, and SQLite database.*

Keywords: *Supply chain optimization, Internet of Things, Machine Learning, inventory management, demand forecasting, and automated report generation.*

I. Introduction: Supply chain optimisation stands as a cornerstone for the success of businesses, and this project endeavours to revolutionize its efficiency through the synergistic integration of Internet of Things (IoT) and Machine Learning (ML) technologies. Recognizing the pivotal role of inventory management in the supply chain optimisation, the project employs IoT for real-time tracking and monitoring, ensuring a dynamic and responsive approach to inventory control. ML algorithms facilitate proactive decision-making and minimizing instances of stockouts or overstock by predicting accurate feature demand using historical data [1] (Figure 1). Beyond conventional inventory management, the project introduces a comprehensive

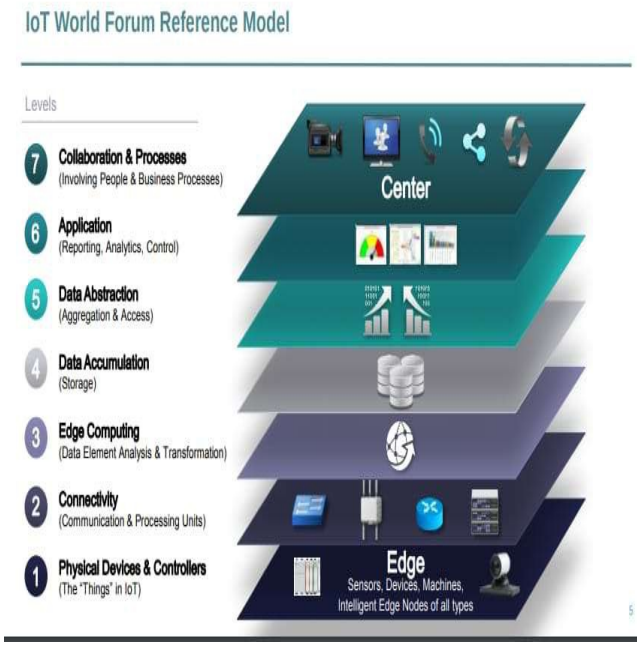


Figure 1: Architecture of Internet of Things.

medicine database, enriching the system with vital details such as uses, side effects, dosage instructions, and necessary precautions. This in maintaining accurate stock levels and also ensures compliance with regulatory standards and enhances overall safety in pharmaceutical supply chains. Furthermore, the integration of environmental temperature monitoring, facilitated by temperature sensors, adds a layer of quality control. This feature is particularly crucial in industries where temperature-sensitive goods, such as pharmaceuticals, require stringent monitoring to preserve their efficacy.In health-sector, the system offers a centralized repository of medicine details, encompassing uses, side effects, dosage instructions, and precautions. Users can easily retrieve essential information, promoting safe and informed medication practices. Additionally, the system utilizes temperature sensors to monitor environmental conditions, ensuring the unity of temperature-sensitive goods during transportation and storage [2].The unification of Django

II. Literature Survey:

and SQLite database enhances data management and system scalability, providing a robust and secure foundation. This interdisciplinary approach harnesses the power of technology to create a seamless and intelligent supply chain ecosystem, fostering efficiency, accuracy, and transparency across inventory and medicine management. The proposed system represents a significant step forward in leveraging technology for optimizing supply chain activities and healthcare practices. In summary, the project envisions a holistic optimization of the supply chain, leveraging IoT and ML for intelligent inventory management, supplemented by a robust medicine database and environmental monitoring [3]. The amalgamation of these elements promises a more resilient, efficient, and responsive supply chain ecosystem, addressing critical facets of modern business challenges [4].

The paper traverses the role of Internet of Things, Machine learning and its impact on supply chain optimisation (SCO) through an extensive literature evaluation. Important aspects of IoT and ML in SCO are covered including IoT definition, main IoT technology enablers, ML algorithms used and various SCO processes and applications. This paper consists of the extant literature, such as based on methodology, industry sector and focus on a classification based on major supply chain processes.

Saeid Sadeghi et al [5] is introducing Machine Learning in Sustainable Industrial Development. The methodology involves a literature analysis of machine learning (ML) techniques applied in supply chain management (SCO), comparing traditional and AI methods, and developing a comprehensive framework for ML applications. It comprises of objectives such as evaluate the efficiency of traditional versus AI methods in handling big data for supply chain management, review and classify commonly used AI methods in different SCO areas and develop a detailed framework for ML applications in supplier selection, risk prediction, and various SCO aspects.

Kamalendu Pal et al [6] employs a literature analysis approach to explore the impact of IoT in supply chain management, emphasizing on sensing enterprises. It comprises of investigate the part of IoT in creating sensing enterprises for supply chain optimisation, examine the layered architecture of supply chain information systems and explore the characteristics of sensing enterprises and their impact on business processes. Future research could focus on Research on data protection in IoT to address both the factors security and privacy concerns. Integration of emerging technologies for further enhancing the process automation. Exploration of novel applications and improvements in sensing enterprise concepts for evolving business needs.

Gian Antonio Susto et al [7] introduces a Multiple Classifier Predictive Maintenance (MC PdM) system for integral faults in semiconductor manufacturing. It utilizes parallel ML classifiers with dynamic decision rules for maintenance management, addressing high-dimensional and censored data issues. It comprises of develop a PdM methodology for integral faults in semiconductor manufacturing, employ multiple classifiers with different prediction horizons for enhanced decision-making and demonstrate the effectiveness using a simulated example and a benchmark semiconductor maintenance problem.

Mikkilineni et al [8] is introducing Treatment Algorithm, and Prevention for the problem statement. A literature explores using PubMed, Ovid MEDLINE and Embase databases was performed with no date restrictions. Search terms included “hyaluronic acid and hypersensitivity” and “hyaluronic acid and nodules.” Data analysed included study type, number of subjects, HA filler type, injection location, adverse reaction, timing, treatment, and outcomes. The goal of this article is to review the hypersensitivity reactions to be reported, in combination with the authors' experience, to create a classification system to sort the timing and clinical manifestations of these reactions, even a treatment schema to manage their clinical course.

Mustafa Abdullah Azzawi et al [9] employed a literature review on IoT healthcare usage and proposed a CoAP-based authentication mechanism using ECC algorithms for IoT devices. The main goals were to enhance healthcare IoT security through a resource-efficient authentication mechanism and assess the viability of CoAP with ECC algorithms. The employed methods offer efficient authentication, utilizing small ECC keys to ensure powerful encryption by decreasing computational requirements, catering to the constraints of IoT devices. Future research could focus on refining the proposed authentication approach, addressing emerging IoT security threats, and exploring broader applications beyond healthcare, ensuring continuous adaptation to evolving technologies.

Xiaoyun yue et al [10] is introducing methods optimise of supply chain based on fuzzy decision-making model in internet of things. customers and to assess the extent to which IoT-enabled process integration impacts supply chain performance and contributes to firm sustainability.

The paper applies models to optimize supply chain operations in the IoT context, addressing uncertainties, introducing six-point fuzzy numbers, and validating the model through simulation for theoretical decision-making support. To address the challenges and uncertainties introduced by Internet of things in SCO. To develop a fuzzy decision-making model for supply chain enterprises under IoT influence. To optimize manufacturing resource allocation strategies within the IoT framework. To minimize overall inventory costs while ensuring satisfactory order fulfilment rates. To provide theoretical support for decision-making in the IoT-enabled supply chain.

Cai shousong1 et al [11] is introducing Revenue Model of SCO. This paper involves the study of optimization of a multi-product supply chain using optimization with Internet of Things technology. The author’s revenue model in two-stage multi-product supply chain operations to analyse the effect of IoT technology on overall supply chain revenue. The imperialist competitive algorithm is introduced and improved to solve the model, and computer simulation optimization methods are utilized to solve examples and compare supply chain revenue before and after implementing IoT technology. It contains supply chain revenue model in a two-stage multi-product supply chain. To solve the optimization model it uses imperialist competitive algorithm. To apply computer simulation optimization methods to solve examples and compare supply chain revenue with and without IoT technology.

Tharaka de Vass et al [12] is introducing Internet of things in SCO: a narrative on retail sector sustainability. It consists of conducting twelve semi-structured interviews with managers and qualitative research approach from the Australian retail industry. Using NVivo, the analysis is done to extract insights from the interview data. The main objectives are to investigate how IoT implementation integrates logistics processes within an organization and externally with suppliers and

Mohd Javaid et al [13] is introducing Sustainable Operations and Computers. The authors have extensively reviewed and critically assessed various sources, including research papers, blogs, and books related to Industry 4.0, sustainability, the environment, and other relevant topics. The comprehensive overview of the subject is synthesized the existing knowledge and identifying gaps in the literature analysis. The research objectives are clearly defined which includes to study significant benefits of Industry 4.0 for sustainable manufacturing, to identify tools and elements of Industry 4.0 for developing environmental sustainability, to study Industry 4.0 processes towards Environmental Sustainability, to study major developments through Industry 4.0 to create a sustainable environment and to identify and study Industry 4.0 applications for developing a sustainable environment.

Abderahman Rejeb et al [14] employs an analysis using the Scopus database to systematically evaluate the development in of supply chain management (SCO) and logistics using Internet of things research within the context. Around 800 and more journal articles published and selected in two-decade period for analysis. The bibliometric parameters include authors, institutions, sources, year of publication, and institutions. The keyword co-occurrence network was introduced in relevant literature. The main objectives are to review and analyse the growth of IoT literature in SCO and logistics, identify key contributors, including journals, authors, and institutions, explore the main countries contributing to academic internet of things literature in SCO and logistics, investigate current trends, hot research topics, and discussions in the reviewed literature and provide researchers with perceptions into internet of things research in SCO and logistics and identify knowledge gaps.

Baha M. Mohsen. [15] has provided the methodology for exploring the developments of digital technologies related to SCO involves literature evaluation, analysis and case studies. The research will delve into existing studies, scholarly articles, and industry reports to gather the knowledge of digital technologies in SCO. The role of digital technology in optimizing SCO is to understand how various digital technologies such as AI, big data, blockchain, and Internet of Things contribute to the optimization of SCO processes, to examine the influence of digital supply chain operations on decision-making: This objective aims to explore how digital strategies enhance decision-making regarding supply and demand, even the interdependencies within the process, to critically assess the impact of digital technologies on businesses and consumers. The research will evaluate how the advent of digital technology has affected business operations, consumer behaviours, and the overall economy and to investigate the unification of digital technologies in performance measurement: This objective involves studying how digital technologies contribute to performance measurement tools, focusing on transparency, efficiency, and familiarity with the process.

III. Problem Statement: The problem at hand is to optimize the pharmaceutical supply chain by leveraging IoT for medicine authenticity checks and Machine Learning for side effect prediction thereby addressing the identified gaps.

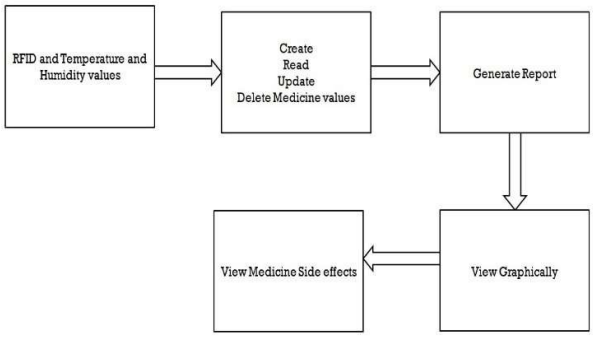
IV. Proposed Methodology: The proposed methodology dwells into different levels which is explained further. The first thing is to RFID integration which incorporate RFID technology into the Django application involves utilizing RFID hardware along with appropriate libraries for interfacing. The process begins with setting up the necessary hardware components and ensuring compatibility with the Django framework. Through RFID readers, the application will be able to capture unique identifiers associated with medicine products. This captured data will then be integrated into the Django backend, where it will be processed and associated

with respective medicine products during addition. The second thing comes CRUD Operations to facilitate the management of medicine products within the Django application, a sequence of CRUD (Create, Read, Update, Delete) operations will be implemented. This involves designing Django models that accurately represent the

1. Django for Web Application Development: Django, a high-level Python web framework, is used for developing the web application. It simplifies the development process with a clean and pragmatic design, aiding in the creation of a strong SCO interface. Django includes features such as an ORM (Object-Relational Mapping) system, authentication, and built-in admin panels, contributing to regulation of **attributes of medicine products, such as name, quanitty with** the application.

its side effects. Through the application's user interface, users will have the ability to perform these CRUD operations. They can create new medicine products, retrieve existing product information, update quantities or other attributes, and delete products as needed. Then comes product availability report which generates a product availability report gathers relevant data from the Django database, including information on each medicine product's name and quantity. This data is then processed to calculate the total quantity available for each medicine product. The report will be presented in a user-friendly format, allowing users to easily assess the availability status of each product. The report gives perception into stock levels, enabling informed decision-making regarding inventory management and procurement and the last thing is side effects information which integrates with external sources, such as APIs or databases containing side effects information for various medicines, is crucial for enhancing the application's functionality. Through this integration, the Django application will be able to retrieve comprehensive side effects information for each medicine product. This information will then be displayed to users within the application, offering perceptions for specific medications. By providing this data, the application enables users to make decisions regarding medication usage with its side effects.

Block Diagram and Description:



2. Python for ML Algorithms: Python is used to implement ML algorithms for demand forecasting. Libraries like scikit-learn and TensorFlow can be utilized to create predictive models based on historical data.ML algorithms enhance the system's ability to anticipate future demand patterns, optimizing inventory levels and improving overall supply chain efficiency.

3. HTML, CSS, and Vanilla JS for Frontend Development: HTML provides the structure, CSS adds styling, and Vanilla JS (JavaScript **without additinoal frameworks) handles client-side interactivity. This** combination ensures a responsive and user-friendly interface for the SCO web application. Frontend development focuses on presenting real-time data, interactive dashboards, and user-friendly features.

4. SQLite Database for Data Storage: SQLite DBMS is chosen because of its simplicity and ease of unification. It stores data related to inventory, medicine details, and environmental monitoring. SQLite provides a structured way to manage and retrieve information critical for supply chain decision-making.

Hardware Requirements:

1. RFID Technology for Inventory Tracking: RFID tags are employed to uniquely identify and track individual items in the inventory. RFID readers gathers data from the tags, providing real-time information on the location and status of each item. This technology facilitates efficient and accurate inventory tracking, reducing manual errors and improving overall supply chain process visibility.

Figure 3: RFID TAG

Figure 2: Block Diagram.

The RFID readers collects data from RFID tags attached to inventory items. The NODEMCU (Internet of Things) facilitates the connection between RFID reader and the central system for real-time data transmission. The data processing and validation module processes and validates RFID data before storing it in the inventory database. The Inventory Database which is SQLite database stores data related to inventory items, ensuring efficient retrieval and updates. Temperature Sensors continuous monitoring of environmental conditions using temperature sensors. Machine Learning Models are Python-based ML models analyse historical data for demand forecasting. Automated Reporting Module generates reports based on inventory status, demand forecasts, and environmental conditions and the Frontend Interface (HTML, CSS, JS) are user-friendly web interface for stakeholders to interact with and visualize the data.

V. System Requirements:

Software Requirements:

2. NODEMCU for IoT Integration: NODEMCU, the ESP8266 module, is utilized to connect devices to the internet physically. It facilitates IoT integration by collecting and sending data from RFID readers and temperature sensors to the central system. NODEMCU ensures seamless communication between hardware components and the web application, enabling real-time monitoring and control.

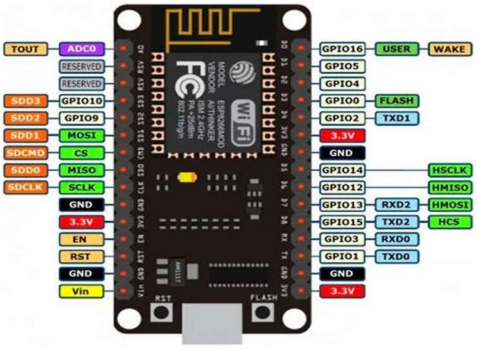


Figure 4: NodeMCU

3. Temperature Sensors

Temperature sensors, such as DS18B20, are employed to monitor environmental conditions. These temperature data, ensuring that temperature sensitive goods, like pharmaceuticals, are stored within specified ranges. Environmental monitoring enhances product quality, compliance, and overall supply chain integrity.

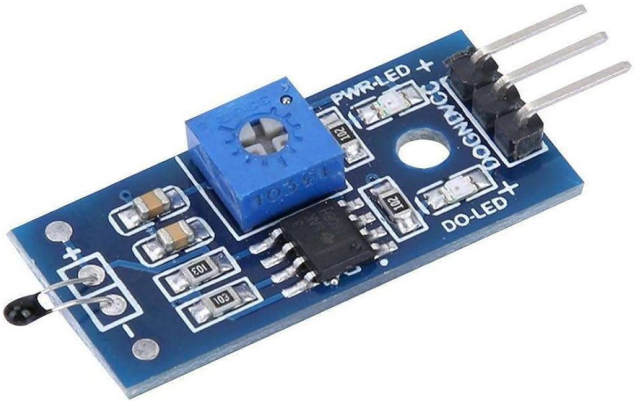


Figure 5: Temperature Sensor

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VII. Results and Discussion:

RFID Integration: Incorporating RFID technology into the Django application involves utilizing RFID hardware along with appropriate libraries for interfacing. The process begins with setting up the necessary hardware components and ensuring compatibility with the Django framework. Through RFID reader, the application will be able to collect unique identifiers associated with medicine products. This collected data will then be unified into the Django backend, where it will be processed and associated with respective medicine products during addition.

Figure 7: Input values from sensor.

CRUD Operations: To facilitate the management of medicine products within the Django application, a sequence of CRUD (Create, Read, Update, Delete) operations will be implemented. This involves

designing Django models that accurately represent the attributes of medicine

VI. Working:

Architecture Diagram and Description:

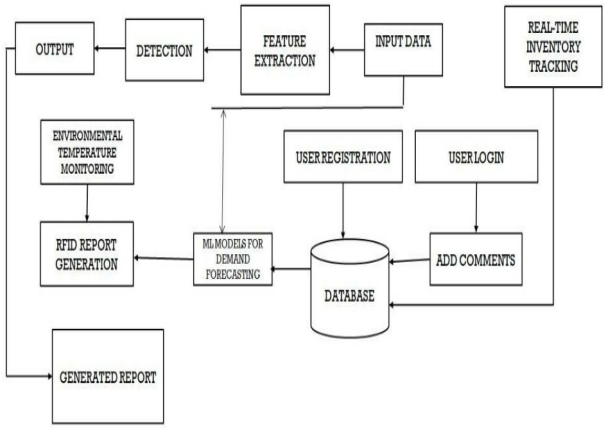

Figure 6: Architecture Diagram.

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the application's user interface, the user will be able to perform these CRUD operations. They can create new medicine products, retrieve existing product information, update quantities or other attributes, and delete products as needed.

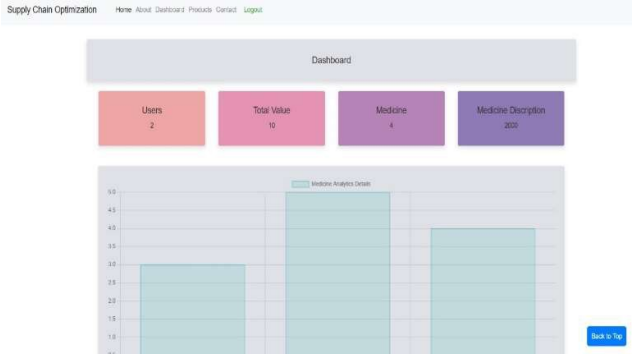

Figure 8: Drug dashboard.

Product Availability Report: Generating a product availability rpeort involves collecting and unifying the applicable data from the Django database, including information on each medicine product's name and quantity. This data is then processed to calculate the total quantity available for each medicine product. The report will be presented in a user-friendly format, allowing users to easily assess the availability status of each product. This report provides valuable insights into stock

levels, enabling informed decision-making regarding inventory management and procurement.

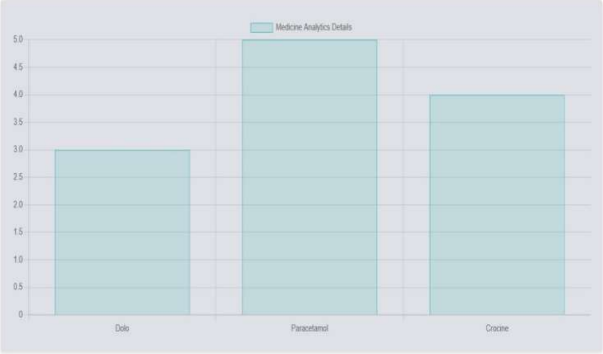

Figure 8: Drug analysis.

Side Effects Information: Integrating with external sources, such as APIs or databases containing side effects information for various medicines, is crucial for enhancing the application's functionality. Through this integration, the Django application will be able to retrieve comprehensive side effects information for each medicine product. This information will then be displayed to users within the application, offering valuable insights into potential risks associated with specific medications. By providing this data, the application aims to empower users with knowledge to make informed decisions regarding medication usage and management of potential side effects.

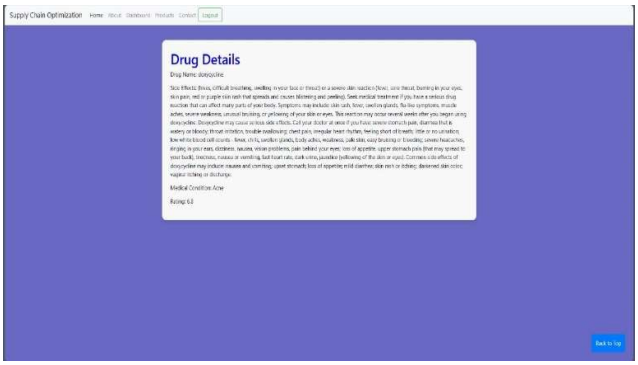

Figure 8: Drug Details.

use the computational facilities by providing us most suitable environment for investigation, research and development. Special thanks to the faculty members of S G Balekundri Institute of Technology, Belagavi, Karnataka, India for their guidance and support.

CONFLICTS OF INTEREST STATEMENTS

The authors declare that there is no conflict of interest in this paper.

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VIII.Conclusion:

RFID (Radio Frequency Identification) technology has emerged as a promising solution for optimizing supply chain management in the pharmaceutical industry. By leveraging RFID tags attached to medicine packages, containers, or pallets, companies can achieve enhanced visibility and traceability throughout the entire supply chain process. This technology enables real-time tracking of inventory, improves accuracy in inventory management, reduces the risk of counterfeit drugs, and enhances overall operational efficiency.One significant aspect of RFID-based supply chain optimization is its potential to address medicine side effects and ensure patient safety. With RFID tags containing detailed information about each medicine and potential side effects, healthcare providers can access accurate and up-to-date information to make informed decisions. This ensures that patients receive the correct medication, reducing the risk of adverse reactions or complications due to medication errors. In conclusion, RFID-based supply chain optimization offers significant benefits for the pharmaceutical industry, includi management and enhanced patient safety.

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